

Phishing Detection and Awareness Report

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Date: April 2026

Course/Program: Future Interns Cybersecurity Task 2

Introduction

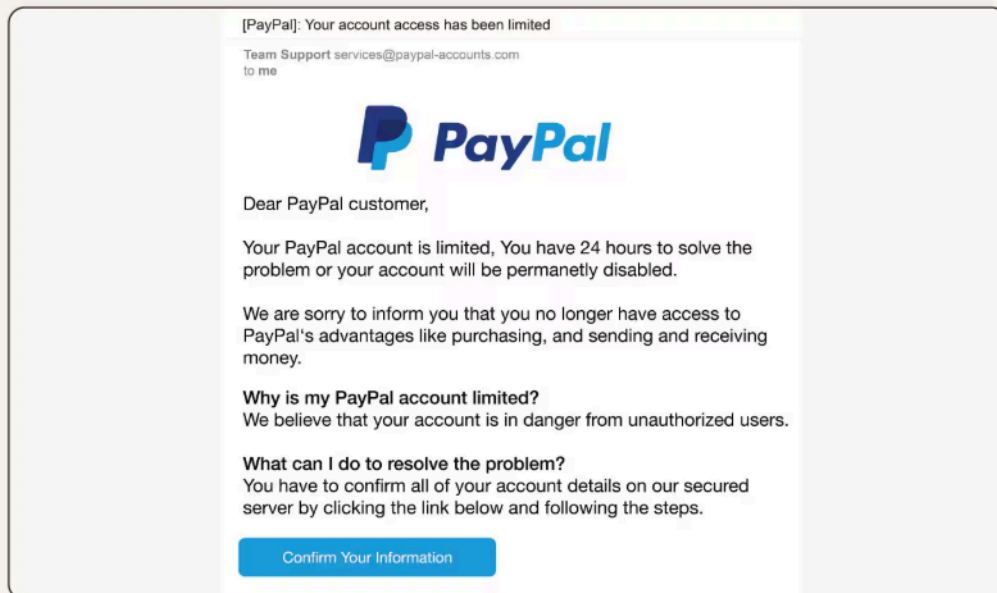
Phishing is a type of cyber attack where criminals send fake emails pretending to be trusted organizations or people - such as banks, tech companies or even colleagues - in order to trick victims into revealing sensitive information, transferring money, or clicking malicious links.

According to cybersecurity research, phishing remains one of the most common and costly cyber threats facing businesses today. Most successful phishing attacks do not happen because of weak technology, they happen because employees do not know how to identify a fake email.

This report analyzes two real-world phishing email samples, identifies their key warning signs, classifies their risk level, and provides clear prevention guidelines that employees and organizations can use to protect themselves from phishing attacks.

Email analysis and how it works

Email sample #1 Fake PayPal Email



Subject Line: Your account access has been limited

Sender Email: services@paypal-accounts.com

Greeting: Dear PayPal customer

What It Wants: Victim to confirm personal information

Urgency Used: Yes — account will be permanently disabled in 24 hours

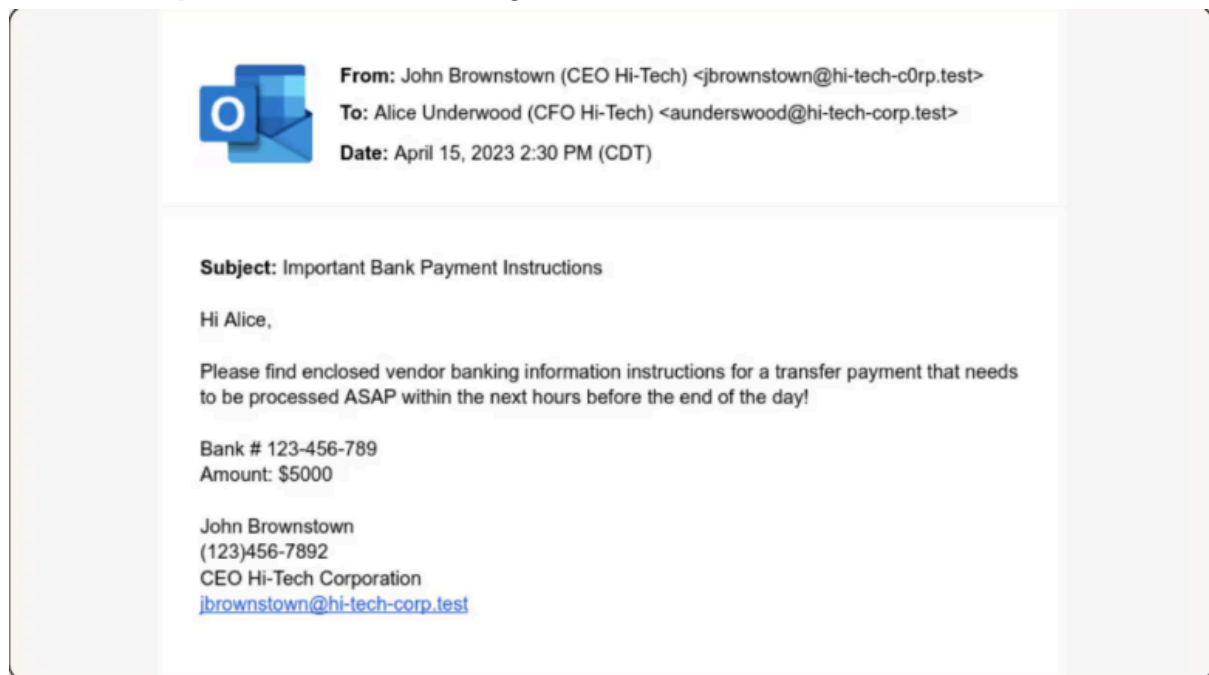
Fake Domain: paypal-accounts.com is not PayPal's real domain (paypal.com)

Risk Level: ● PHISHING

How it works:

This email pretends to be from PayPal to trick the user into thinking their account has a problem. It creates panic by threatening to permanently disable the account within 24 hours. The goal is to rush the user into clicking a link that leads to a fake PayPal website. Once there, the user is asked to enter their login details or personal information, which the attacker then steals. The sender's email domain — paypal-accounts.com — is fake. PayPal's real domain is simply paypal.com

Email sample #2 Fake Bank Payment Email



Subject Line: Important Bank Payment Instructions

Sender Email: jbrownstown@hi-tech-c0rp.test

Greeting: Hi Alice

What It Wants: Victim to urgently transfer money

Urgency Used: Yes — ASAP

Fake Domain: hi-tech-c0rp.test uses zero instead of letter O in "corp"

Risk Level: ● PHISHING

How it works:

This email pretends to be a legitimate business payment instruction, likely impersonating a manager or finance contact named 'J Brownstown.' It addresses the victim by name (Alice) to appear more convincing and trustworthy. The attacker creates urgency by demanding the transfer happen 'ASAP', leaving the victim no time to think or verify. The sender's domain — hi-tech-c0rp.test — is suspicious because it replaces the letter O with a zero, a common trick used to make fake domains look real at a quick glance. This type of attack is also known as Business Email Compromise (BEC).

Risk Classification

Each analyzed email has been classified according to the following risk scale:

Risk Level	Meaning
 Safe	Legitimate email, no suspicious indicators
 Suspicious	Some warning signs but not fully confirmed as phishing
 Phishing	Clearly fake email designed to deceive the victim

Email Sample	Subject Line	Sender Domain	Classification
Sample #1	Your account access has been limited	paypal-accounts.com	 Phishing
Sample #2	Important Bank Payment Instructions	hi-tech-c0rp.test	 Phishing

Both emails were classified as Phishing based on the following reasons:

- Use of fake sender domains impersonating trusted organizations
- Urgent and threatening language designed to pressure the victim
- Requests for sensitive actions such as confirming personal information or transferring money
- Generic or manipulative greetings intended to appear legitimate

Prevention Tips

The following tips will help employees and individuals protect themselves from phishing attacks:

1. Always check the sender's email address carefully before responding or clicking any links. Look out for slight misspellings or unusual domains.
2. Never click on links or download attachments from emails you were not expecting, even if they appear to come from a trusted source.
3. If an email asks you to confirm personal information, log in directly to the official website by typing the address in your browser instead of clicking any link in the email.
4. Be suspicious of any email that creates urgency or threatens negative consequences such as account suspension or legal action.
5. When in doubt, contact the supposed sender directly through official channels such as a phone number from their official website to verify the request.
6. Always report suspicious emails to your IT or security team immediately instead of ignoring or deleting them.
7. Keep your email security software and antivirus up to date to help detect and block phishing attempts automatically.

Do's and Don'ts for Employees

DO's

Do verify the sender's email address before responding to any email.

Do go directly to the official website by typing the address in your browser instead of clicking email links.

Do report any suspicious emails to your IT or security team immediately.

Do take your time before acting on any email, especially those demanding urgent action.

Do use strong and unique passwords for all your accounts and enable two factor authentication where possible.

Don'ts

Do not click on links or download attachments from unexpected or suspicious emails.

Do not share your password, OTP, or any personal information via email.

Do not transfer money based on email instructions alone without verifying through official channels.

Do not ignore suspicious emails — always report them.

Do not assume an email is safe just because it uses a company logo or familiar name.

Conclusion

Phishing attacks remain one of the most serious cybersecurity threats facing individuals and organizations today. As demonstrated by the two email samples analyzed in this report, attackers use clever tricks such as fake domains, urgent language, and impersonation to deceive victims into revealing sensitive information or transferring money.

The good news is that phishing attacks are largely preventable. With the right knowledge and awareness, employees can learn to identify suspicious emails before falling victim to them. Simple habits such as checking sender domains carefully, avoiding clicking unexpected links, and verifying urgent requests through official channels can make a huge difference in keeping both individuals and organizations safe.

This report serves as a practical awareness guide that organizations can use to educate their employees about phishing threats and how to respond to them effectively. Cybersecurity is not just the responsibility of IT teams — it is everyone's responsibility.